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8

Conclusion

STUDIES OF HOW TECHNOLOGY interacts with the international landscape often fixate on the most dramatic aspect of technological change—the eureka moment. Consistent with this frame, the standard explanation for the technological causes of economic power transitions emphasizes a rising power’s ability to dominate profits in leading sectors by generating the first implementation of radical inventions. This book draws attention, in contrast, to the often unassuming process by which an innovation spreads throughout an economy. The rate and scope of diffusion is particularly relevant for GPTs—fundamental advances like electricity or AI that have the potential to drive pervasive transformation across many economic sectors.

Based on the process of GPT diffusion, this book puts forward an alternative theory of how and when significant technological breakthroughs generate differential rates of economic growth among great powers. When evaluating how technological revolutions affect economic power transitions, GPTs stand out as historical engines of growth that can provide major boosts to national productivity. Though each is different, GPTs tend to follow a common pattern: after multiple decades of complementary innovations and institutional adaptations, they gradually diffuse across a broad range of industries. Everything, everywhere, but not all at once.

This impact pathway markedly diverges from existing theories based on leading sectors. Akin to a sprint on a narrow lane, great power competition over leading sectors is framed as a race to dominate initial breakthroughs in the early growth periods of new industries. In contrast, GPT diffusion theory proposes that by more effectively adopting GPTs across many application sectors, some great powers can sustain higher levels of productivity growth than their competitors. Like a marathon on a wide road, great power competition over GPTs is a test of endurance.

Disruptive technological advances can bring about economic power transitions because some countries are more successful at GPT diffusion than others. A nation's effectiveness at adapting to emerging technologies is determined by the fit between its institutions and the demands of those technologies. Thus, if economic power transitions are driven by the GPT trajectory, as opposed to LS product cycles, the institutional adaptations that matter most are those that facilitate information exchanges between the GPT sector and application sectors, in particular the ability of nations to widen the engineering skill base linked to a new GPT.

Three historical case studies, designed and conducted in a way to assess the explanatory power of the GPT mechanism against the LS mechanism, provide support for GPT diffusion theory. The case studies cover periods characterized by both remarkable technological change—the “three great industrial revolutions” in the eyes of some scholars—and major fluctuations in the global balance of economic power.¹ Overall, the case study evidence underscores the significance of GPT diffusion as the key pathway by which the technological changes associated with each industrial revolution translated into differential rates of economic growth among the great powers.

In the case of Britain's rise to economic preeminence during the First Industrial Revolution, expanded uses of iron in machine-making spurred mechanization, the key GPT trajectory. The gradual progression of mechanization aligned with the period when Britain's productivity growth outpaced that of France and the Netherlands. Britain's proficiency in adopting iron machinery across a wide range of economic activities, rather than export advantages from dominating innovation in leading sectors such as cotton textiles, proved more central to its industrial ascent. Though its industrial rivals boasted superior systems of higher technical education for training expert scientists and top-flight engineers, Britain benefited from mechanics' institutes, educational centers like the Manchester College of Arts and Sciences, and other associations that expanded access to technical literacy and applied mechanics knowledge.

The Second Industrial Revolution case also favors the GPT mechanism's explanation of why certain great powers better adapt to periods of remarkable technological change. The LS mechanism focuses on Germany's discoveries in new science-based industries, such as chemicals, as the driving force behind its catching up to Britain before World War I. However, the United States, emerging as the preeminent economic power during this period, was more

1. Von Tunzelmann 1997, 2.

successful than Germany in exploiting the technological opportunities of the Second Industrial Revolution. Enabled by innovations in machine tools, the extension of interchangeable manufacturing techniques across many American industries functioned as the key GPT trajectory that fueled the rise of the United States. Scientific infrastructure or industrial R&D capabilities, areas in which the United States lagged behind its industrial rivals, cannot account for its advantage in adopting special-purpose machinery across nearly all branches of industry. Rather, the United States gained from institutional adaptations to widen the base of mechanical engineering talent, including through the expansion of technical higher education schools and the professionalization of mechanical engineering.

Evidence from the US-Japan rivalry amid the information revolution exposes more gaps in the LS account. During the late twentieth century Japan captured global market shares in new fast-growing sectors such as consumer electronics and semiconductor components, prompting many to predict that it would overtake the United States as the leading economic power. Yet such an economic power transition, an inevitability based on the expectations of the LS mechanism, never occurred. Instead, the United States sustained higher rates of economic growth than Japan owing, in part, to greater spread of computerization across many economic sectors. Japan's productivity growth kept up with the US rate in sectors that produced information technology but lagged far behind in sectors that intensively used information technology. Once again, differences in institutional adaptations to widen the GPT skill base turned out to be significant. While Japanese universities were very slow to adapt their training to the demand for more software engineers, US institutions effectively broadened the pool of such skills by cultivating a separate discipline of computer science.

As a supplement to the case studies, I conducted a large-*n* statistical analysis to test whether countries with superior GPT skill infrastructure preside over higher rates of GPT diffusion. Leveraging time-series cross-sectional data on software engineering education and computerization rates in nineteen countries (the G20 economies) across twenty-five years, the quantitative analysis confirmed this crucial expectation derived from GPT diffusion theory. I found less support for other factors often assumed to have a positive effect on an economy's broader technological transformation, including institutional factors linked to securing LS product cycles. This empirical test validates a core component of GPT diffusion theory across a sample of the world's major emerging and developed economies.

Main Contributions

First, at its core, *Technology and the Rise of Great Powers* introduces and defends GPT diffusion theory as a novel explanation for how and when technological change can lead to a power transition. Historical case studies and statistical analysis substantiate the explanatory power of GPT diffusion theory over the standard explanation of technology-driven power transitions based on leading sectors, which exerts enduring influence in policy and academic circles.² In doing so, the book answers the call by scholars such as Michael Beckley and Matthew Kroenig for the international relations field to devote more attention to the causes of power transitions, not just their consequences.³

By expounding on the significance of GPT skill infrastructure, the book points toward next steps to better understanding the politics behind some of the most significant technological advances in human history. To investigate why some countries are more successful in cultivating GPT skill infrastructure, promising avenues of research could tap into existing work that concentrates on centralization, inclusiveness of political institutions, government capacity to adopt long time horizons, and industrial organization.⁴ In these efforts, being careful to differentiate between various pathways by which technological changes make their mark, as this book does with the GPT and LS mechanisms, will be especially important when underlying political factors that satisfy the demands of one type of technological trajectory run counter to the demands of another.

Future work should also probe other institutional factors beyond GPT skill infrastructure that contribute to cross-national differences in GPT adoption. This opens up a universe of institutions that are often ignored in innovation-centered accounts of technological leadership, including gender gaps in engineering education,⁵ transnational ethnic networks that facilitate technology transfer,⁶ and “technology diffusion institutions,” such as standard-setting organizations and applied technology centers.⁷

2. Drezner 2019, 289; Kennedy 2018; Tellis et al. 2000.

3. Beckley 2021; Kroenig 2020.

4. Acemoglu and Robinson 2006, 2012; Doner and Schneider 2016; Simmons 2016.

5. On changing gender norms and mechanics' education in the First Industrial Revolution, see Jacob 1997, 211.

6. Taylor 2016; Saxenian and Hsu 2001.

7. Shapira and Youtie 2017.

In positioning the LS mechanism as the main foil to the GPT mechanism, my intention is to use this clash between theories to productively advance our understanding of the rise and fall of great technologies and powers. Contestation should not be misread as disparagement. In one sense, GPT diffusion theory builds on previous scholarship about leading sectors, which first identified the need to flesh out more specific linkages between certain technological advances and more highly aggregated economic changes in the context of power transitions.⁸ Testing, revising, and improving upon established theories is essential to gradual yet impactful scientific progress—not so unlike the incremental, protracted advance of a GPT.

Second, the book's central argument also suggests revisions to assessments of power in international politics. Recognizing that scientific and technological capabilities are becoming increasingly central to a nation's overall power, researchers tend to equate technological leadership with a country's ability to initiate "key 'leading sectors' that are most likely to dominate the world economy into the twenty-first century."⁹ For instance, an influential RAND report, "Measuring National Power in the Postindustrial Age," proposes a template for measuring national power based on a country's capacity to dominate innovation cycles in "leading sectors."¹⁰ In this effort, the authors draw directly from LS-based scholarship: "The conceptual underpinnings of this template are inspired by the work of Schumpeter, Rostow, Gilpin, Kennedy, and Modelski and Thompson."¹¹ This study has gained considerable traction in academic and policymaking circles, inspired further workshops on national power, and has been called "the definitive US study on CNP [Comprehensive National Power]."¹²

Contrary to these approaches, this book submits that evaluations of scientific and technological power should take diffusion seriously. Assessments that solely rely on indicators of innovation capacity in leading sectors will be misleading, especially if a state lags behind in its ability to spread and embed innovations across productive processes. A more balanced judgement of a state's potential for technological leadership requires looking beyond multinational corporations, innovation clusters like Silicon Valley, and eye-popping R&D

8. Thompson 1990, 221.

9. Wohlforth 1999, 17.

10. Tellis et al. 2000, 37.

11. *Ibid.*, 36. In the order cited in the quote, see Schumpeter 1934, 1939; Rostow 1960; Gilpin 1981; Kennedy 1987; and Modelski and Thompson 1996.

12. Singh, Gera, and Dewan 2013, 60; Treverton and Jones 2005.

numbers to the humble undertaking of diffusion. It shines the spotlight on a different cast of characters: medium-sized firms in small towns, engineers who tweak and implement new methods, and channels that connect the technological frontier with the rest of the economy.

In an article published in the *Review of International Political Economy* journal, I illustrated the value of this diffusion-oriented approach in gauging China's scientific and technological capabilities.¹³ Preoccupied with China's growing strength in developing new-to-the-world advances, existing scholarship warns that China is poised to overtake the United States in technological leadership. This is mistaken. There is still a large gap between the United States and China when it comes to the countries' readiness to effectively spread and utilize cutting-edge technologies, as measured by penetration rates of digital technologies such as cloud computing, smart sensors, and key industrial software. When the focus shifts away from impressive and flashy R&D achievements and highly cited publications, China's "diffusion deficit" comes to light. Indeed, a diffusion-centric assessment indicates that China is much less likely to become a scientific and technological superpower than innovation-centric assessments predict.

Relatedly, the GPT diffusion framework can be fruitfully applied to debates about the effects of emerging technologies on military power. Major theories of military innovation focus on relatively narrow technological developments, such as aircraft carriers, but the most consequential military implications of technological change might come from more fundamental advances like GPTs. In an article that employs evidence from electricity's impact on military effectiveness to analyze how AI could affect the future of warfare, Allan Dafoe and I challenge studies that predict AI will rapidly spread to militaries around the world and narrow gaps in capabilities.¹⁴

Third, as chapter 7 spells out in detail, GPT diffusion theory provides an alternative model for how revolutionary technologies, in particular AI, could affect the US-China power balance. This, in turn, implies different optimal policies for securing technological advantage. Drawing on the LS template, influential thinkers and policymakers in both the United States and China place undue emphasis on three points: the rapid timeframe of economic pay-offs from AI and other emerging technologies; where the initial, fundamental innovations in such technologies cluster; and growth driven by a narrow range of economic sectors.

13. Ding 2023.

14. Ding and Dafoe 2023.

GPT diffusion theory suggests diverging conclusions on all three dimensions. The key technological trajectory is the relative success of the United States and China in adopting AI advances across many industries in a gradual process that will play out over multiple decades. It will be infeasible for one side to cut the other off from foundational innovations in GPTs. The most important institutional factors, therefore, are not R&D infrastructure or training grounds for elite AI scientists but rather those factors that widen the skill base in AI and enmesh AI engineers in cross-cutting networks with entrepreneurs and scientists.¹⁵

Yet, the United States is fixated on dominating innovation cycles in leading sectors. When it comes to their grand AI strategy, US policymakers are engrossed in ensuring that leading-edge innovations do not leak to China, whether by restricting the immigration of Chinese graduate students in advanced technical fields or by imposing export controls on high-end chips for training large models like GPT-3 and ChatGPT.¹⁶ A strategy informed by GPT diffusion theory would, instead, prioritize improving and sustaining the rate at which AI becomes embedded in a wide range of productive processes. For instance, in their analysis of almost 900,000 associate's degree programs, Center for Security and Emerging Technology researchers Diana Gehlhaus and Luke Koslosky identified investment in community and technical colleges as a way to unlock "latent potential" in the US AI talent pipeline.¹⁷ This recommendation accords with an OECD working paper on the beneficial effects of a wider ICT skills pool on digital adoption rates across twenty-five European countries. The study finds that "the marginal benefit of training for adoption is found to be twice as large for low-skilled than for high-skilled workers, suggesting that measures that encourage the training of low-skilled workers are likely to entail a double dividend for productivity and inclusiveness."¹⁸

At the broadest level, this book demonstrates a method to unpack the causal effects of technological change on international politics. International relations scholars persistently appeal for the discipline to better anticipate the

15. I am not arguing that policies and institutional adaptations that nurture cutting-edge innovations are counterproductive. After all, R&D investments have been referred to as "diffusion policies in disguise" because they can also have a positive impact on diffusion, even if that is not their main goal. But why go for the disguise when you can have the real thing? (Stoneman and Diederer 1994; see also Fagerberg 1987; Howitt and Mayer-Foulkes 2002).

16. Hua 2021; Palmer 2023.

17. Gelhaus and Koslosky 2022.

18. Nicoletti, von Rueden, and Andrews 2020.

consequences of scientific and technological change, yet these demands remain unmet. By one measure, between 1990 and 2007, only 0.7 percent of the twenty-one thousand articles published in major international relations journals explicitly dealt with the topic of science and technology.¹⁹ One bottleneck to researching this topic, which Harold Sprout articulated back in 1963, is that most theories either grossly underestimate the implications of technological advances or assume that technological advance is the “master variable” of international politics.²⁰

This book shows that the middle ground can be a place for fruitful inquiry. Technology does not determine the rise and fall of great powers, but some technological trends, like the diffusion of GPTs, do seem to possess momentum of their own. Social and political factors, as represented by GPT skill infrastructure, shape the pace and direction of these technological trajectories. This approach is particularly useful for understanding the effects of technological change across larger scales of time and space.²¹

19. Mayer, Carpes, and Knoblich 2014, 14.

20. Sprout 1963, 187.

21. Dafoe 2015; Herrera 2006; Mayer, Carpes, and Knoblich 2014.

